

Figure 1: The relationship between domain, technology and software expertise

If you are good in technology, then you could be a technology consultant. For example, you could be a Digital Signal Processing (DSP) consultant, a Reliability and Availability consultant, or an Embedded Systems consultant.

Some people are experts at neither technology nor software, but understand the domain very well. If you are a chartered accountant or a doctor of medicine, then you are a domain expert/consultant, and could play the role of a domain consultant in building products based on software.

Some people understand both software and technology; they are called Technical Specialists. Figure 1 shows the relationship between domain, technology and software expertise.

An architect should be an expert in a particular technology, and in at least one area of software development, with a reasonable background in the domain that the product or service is being designed for. Figure 2 shows that an architect can be an expert in software or hardware.

Let's look at an example to help us understand what an architect does.

Let's assume you are an architect who develops products for automobiles. The domain is automobile engineering; the technology may be GPS- (Global Positioning System) based navigation systems using embedded Linux, and the programming languages are C or C++.

Having been an architect in the automotive domain for many years, you cannot become an architect in the Consumer Electronics domain without reasonable experience in this new domain, but you can still be a technology consultant or technical specialist in the Automotive Electronics (AE) or Consumer Electronics (CE) domains—or even switch between them. This is because some technologies overlap between these two domains. Movement across domains as an architect is very difficult, if not impossible. Let us understand the roles of hardware and software architects.

Software and hardware architects

Both software and hardware architects need to have domain knowledge and technology knowledge. Both need hands-on skills in their area of expertise—software expertise and hardware expertise, respectively.

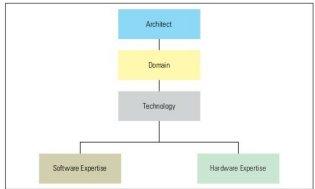


Figure 2: Areas for an architect

However, you will have hands-on skills only if you have worked as a developer on a product, and written many thousands of lines of code; completely built a product or a software feature from scratch, or done board design from start to finish.

If you are an architect at the product, systems or solutions level, you should have done both hardware and software development at some point in your career. Having both hardware and software viewpoints will allow you to visualise the system or solution clearly, and decompose them at the product level. In short, you can easily build the whole from the parts, if you are aware of both hardware and software. Now let's find out if the roles of an architect are slightly different in software services and product organisations.

The role of the architect

Software architects create software that lasts, and their role depends on whether they are working in a software services or a product development organisation.

The role of architects is always the same—they create products or features out of what already exists or something completely new—and they do this by following certain principles of software and hardware development. Modern-day service and product organisations offer different roles to an architect. In a service organisation, an ideal architect:

- Resolves technical issues, or provides solutions for difficult problems, if this cannot be done by any other person on the team.
- Attends weekly status meetings, and provides technical support to the offshore team, enhancing the confidence levels of the client.
- Delegates issues to the project lead or senior engineers, if required.
- Keeps a tab on the technical competency of the team, and helps build it.
- Understands the work preferences, technical knowledge and ethos of the team members, and appropriately helps the Project Manager distribute work among them.
- Is conversant with the evolution of the product and the software.
- Helps the Business Development Manager and the Project